

$$\left(\frac{x^2 - 4}{x^2 + 1} \right)' = \frac{x^2 + 1 \cdot 2x - x^2 - 4 \cdot 2x}{(x^2 + 1)^2}$$

$$\left(\frac{\sin x}{1 + \cos x} \right)' = \frac{1 + \cos x \cdot \cos x - \sin x \cdot -\sin x}{(1 + \cos x)^2}$$

$$= \frac{1 + \cos^2 x + \sin^2 x}{(1 + \cos x)^2}$$